

MORADABAD INSTITUTE OF TECHNOLOGY
Mechanical Engineering Department
 Class Test- I
 (1st Semester) Session-2024-25

Subject Name: Fundamentals of Mechanical Engineering
Subject Code: BME-101

M.M:20
Duration: 1hr

Q.No.	1	2	3	4	5	6
CO	1		2	2	1	1
Bloom's level	K1	K3	K2	K2	K3	K3

Note: 1) This paper contains three sections. Section (A), (B) & (C)

2) All sections are compulsory.

SECTION (A)

Attempt all questions. Each question carries 2 marks. (2*1=2)

1. Explain Modulus of Elasticity and Bulk Modulus.

SECTION (B)

Attempt all questions. Each question carries 3 marks. (3*2=6)

2. Draw stress- strain diagram for a mild steel and explain all salient points on it.

OR

Determine the resultant of force system shown in fig.1

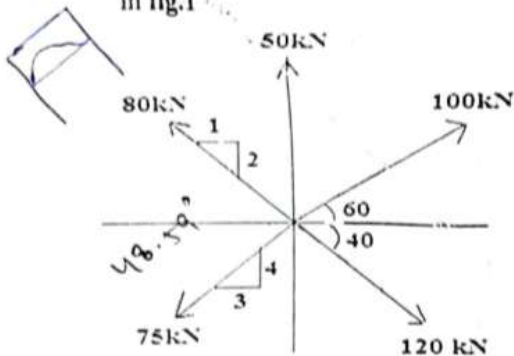


Fig.1

3. Explain the construction of two stroke IC Engine with neat sketch.

SECTION (C)

Attempt all questions. Each question carries 4 marks. (4*3=12)

4. Explain, with neat sketches, the working of a four-stroke SI engine.

5. Determine the reaction at supports of the loaded beam shown in Fig2.

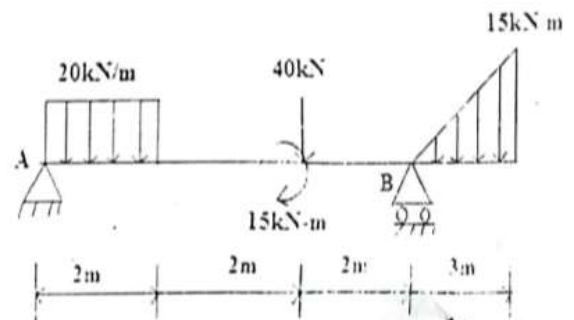


Fig.2

6. Two cylinders of diameters 70 mm and 120 mm weighing 80 N and 120 N respectively are placed in a trough as shown in fig.3. Neglecting friction find the reactions at contact surfaces A, B, C and D

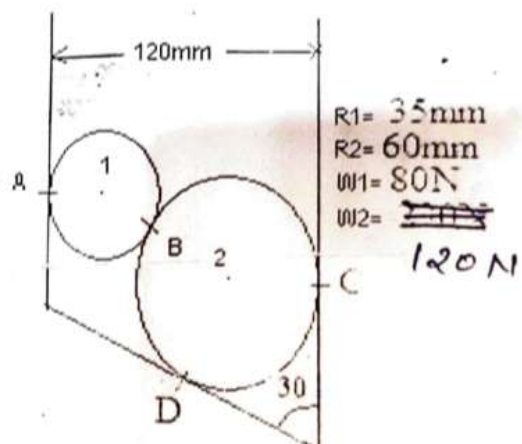


Fig.3.

(R) Sumit
 10/4/11/2024

MORADABAD INSTITUTE OF TECHNOLOGY

Mechanical Engineering Department

Class Test-2

(1st Semester) 2024-25

Set-I

Subject Name: Fundamentals of Mechanical Engineering

Subject Code: BME-101

M.M:20

Duration: 11/1r

Q.No.	1	2	3	4	5	6
CO	4	3	2	3	3	4
Bloom's level	K ₂	K ₂	K ₂	K ₂	K ₃	K ₃

Note: 1) This paper contains three sections. Section (A), (B) & (C).

2) All sections are compulsory.

SECTION (A)

Attempt all questions. Each question carries 2 marks.

(2*1=2)

1. Explain the Newtonian and Non-Newtonian fluids.

SECTION (B)

Attempt all questions. Each question carries 3 marks.

(3*2=6)

2. (a) Explain the following term (i) Relative Humidity (ii) Specific humidity.
(b) Establish the relation between C.O.P of refrigerator and C.O.P of Heat Pump.
3. State the working principle of an electric vehicle. Explain different components of Electric vehicle.

SECTION (C)

Attempt all questions. Each question carries 4 marks.

(4*3=12)

4. Discuss the basic vapour compression refrigeration cycle with T-s and p-h diagram and describe the working of domestic refrigerator.
5. The capacity of a refrigerator is 150TP, when working between -5°C and 25°C. Determine (i) the mass of ice produced per day from water at 25°C. (ii) The power required to drive the unit. Assume that the cycle operates on reversed Carnot cycle and latent heat of ice is 335kJ/kg.
6. (a) Explain the following fluid properties (i) Specific weight (ii) Mass Density (iii) Specific gravity.
(b) Determine the specific gravity of a fluid having viscosity 0.05poise and kinematic viscosity 0.035 stokes.

Set-1

MORADABAD INSTITUTE OF TECHNOLOGY

Mechanical Engineering Department

Makeup Class Test- 1

(1st Semester) Session-2024-25

Subject Name: Fundamentals of Mechanical Engineering
Subject Code: BME-101

M.M:20

Duration: 1hr

Q.No.	1	2	3	4	5	6
CO	1	1	2	2	1	1
Bloom's level	K1	K3	K2	K2	K3	K3

Note: 1) This paper contains three sections. Section (A), (B) & (C)

2) All sections are compulsory.

SECTION (A)

Attempt all questions. Each question carries 2 marks. (2*1=2)

1. Explain the moment of a force and state the varignon's theorem.

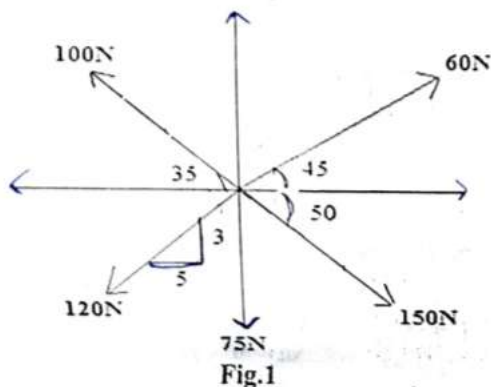
SECTION (B)

Attempt all questions. Each question carries 3 marks. (3*2=6)

2. Derive the relation between modulus of Elasticity, Modulus of Rigidity and Poisson's Ratio.

OR

Determine the resultant of force system shown in fig.1



3. Differentiate between SI and CI engine. Explain TDC, BDC, Swept Volume, Clearance Volume of IC engine

SECTION (C)

Attempt all questions. Each question carries 4 marks. (4*3=12)

4. Explain, with neat sketches, the working of a two-stroke CI engine.

5. Determine the reaction at supports of the loaded beam shown in Fig2.

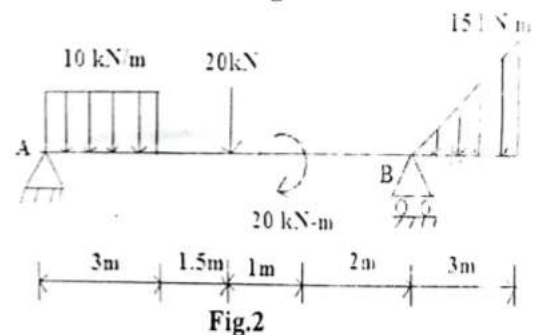


Fig.2

6. Two smooth spheres each of radius 75 mm and weighing 450 N, rest in a horizontal channel having vertical walls as shown in fig.3. Find the reactions at all contact points.

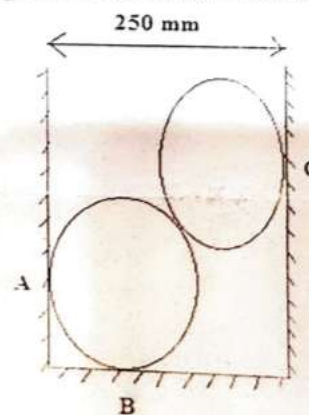


Fig3.

Two
450 N
75 mm
20.75